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Model exam

1. The Majority of the lakes in Ethiopia are found in ...?
 - a. North-East Coast
 - b. Rift Valley Basin
 - c. Nile basin
 - d. Shebelle-Juba basin
2. Which of the following river basin of Ethiopia is transboundary/ non-landlocked ...?
 - a. Awash
 - b. Danakil
 - c. Ogaden
 - d. Mereb
3. Which of the following river basin of Ethiopia has high irrigation potential ...?
 - a. Omo-Gibe
 - b. Genale-Dawa
 - c. Awash
 - d. Blue Nile (Aby)
4. Which of the following river basin of Ethiopia has high hydropower potential ...?
 - a. Omo-Gibe
 - b. Blue Nile (Aby)
 - c. Genale-Dawa
 - d. Awash
5. About 97 % of the total volume of the world's water is estimated to exist as ___ and only 3% as _____.
 - a. Freshwater, oceans
 - b. Groundwater, oceans
 - c. Oceans, freshwater
 - d. None
6. Which of the following is not the reason of water scarcity?

- a. Over-exploitation
 - b. Excessive use
 - c. Growing population
 - d. None of the above
7. Which one of the following is not the reason for the increasing pressure on the world's freshwater resources?
- a. Population growth and pollution
 - b. Urbanization and industrialization
 - c. Decrease withdrawals and climate change
 - d. Increased food production
8. Which of the following is the benefit of rainwater harvesting?
- a. Flood mitigation
 - b. Provide a lot of water to play
 - c. Create good aesthetic view
 - d. Decrease the ground water level
9. Which of the following is not a method for water conservation?
- a. Rainwater harvesting
 - b. Groundwater extraction
 - c. Improving irrigation efficiency
 - d. Avoiding water wastage
10. Which of the following is a water source type that constitutes about 89% of the total fresh water resources in the planet?
- a. Rivers and lakes
 - b. Groundwater
 - c. Rainwater
 - d. Surface water
11. The highest proportion of the total water used in Ethiopia is in which one of the following sectors?
- a. Irrigation
 - b. Industries
 - c. Domestic use
 - d. Ecological and environmental use.
12. The current global climate change is mostly due to?
- a. Volcanic activities

- b. Human activities
 - c. Solar Variations
 - d. Plate tectonics
13. Which of the following is not the cause of warming to the earth's climate?
- a. Increase of Volcanism
 - b. Decrease of chemical weathering
 - c. Respiration
 - d. Increase of chemical weathering
14. Climate change has posed a huge risk to global and regional food security, and the risk is greatly exacerbated in developing countries (most African countries)? As expert, what can you do to minimize this risk in African countries?
- a. Promote mitigation rather than adaptation
 - b. Promote/increase adaptive capacities
 - c. Reduced emission
 - d. B and C are correct
15. We know that orbital variation of the Earth, solar variation, plate tectonic and other factors cause changes in the climate. If we remove the human impacts of greenhouse gas emissions, what might the climate be doing today, on its own?
- a. No change
 - b. Slightly cooling
 - c. Slight warming
 - d. Strong warming
16. Let us say you are working as Green House Gas (GHGs) emission advisory for business community that have many industries, so what feasible method you can advise them to reduce the release GHGs without decreasing the production of these gases?
- a. Adopting new technologies that do not use fossil fuels
 - b. Eliminating industry
 - c. Sequestering
 - d. Mitigating
17. Let us say you have your own private car, bus service of your institution and most importantly your house is about 15 minutes a walking distances from your working institution, so as climate change advocate, what actions could you take to help lessen the impacts of climate change?

- a. Walk
 - b. Use public transport
 - c. Use your own car
 - d. Use institution's bus service
18. Climate change/variability driven drought is quite common in the lowland parts of Ethiopia (like in Afar, Tigray, Wollo and Borena area) and imagine that there are no any resources or living thing in that drought affected areas and what do you do to avoid vulnerability?
- a. Plant tree
 - b. Deliver animal feed and other supports
 - c. Bring food and cloth to the vulnerabel society
 - d. Do nothing
19. As expert of natural resource management, what can you advice to prevent soil erosion and mitigate climate change?
- a. Over growing
 - b. Afforestation
 - c. More livestock and uncontrolled over grazing
 - d. Promote physical soil erosion preventive structure
20. Among the following gasses which one is responsible to climate change?
- a. Oxygen
 - b. Helium
 - c. Carbon dioxide
 - d. Nitrogen
21. Earth would be too cold to support life without _____?
- a. Global warming
 - b. Climate change
 - c. Greenhouse effect
 - d. Climate variability
22. What is the name of the international group of scientists who evaluate scientific studies, and provide climate scenarios data related to any aspect of climate change?
- a. World Meteorological Organization (WMO)
 - b. Montreal Protocol
 - c. Environmental Protection Agency (EPA)
 - d. Intergovernmental Panel on Climate Change (IPCC)

23. What is the primary goal of integrated watershed management?
- a. To maximize agricultural production within the watershed
 - b. To minimize pollution within the watershed
 - c. To balance the social, economic, and **environmental** needs of the watershed
 - d. To control flooding within the watershed
24. What is the first step in the integrated watershed management process?
- a. Flood risk assessment
 - b. **Stakeholder engagement**
 - c. Data collection and **analysis**
 - d. Watershed restoration
25. What role do stakeholders play in integrated watershed management?
- a. They have no role in the process
 - b. They **provide** input and support for decision-making
 - c. They make all the decisions for the watershed
 - d. They are responsible for implementing all management actions
26. Which of the following is an example of non-point source pollution in a watershed?
- a. A factory discharging chemicals directly into a river
 - b. A sewage treatment plant releasing treated water into a river
 - c. **Runoff** from agricultural fields carrying pesticides and fertilizers into a river
 - d. A leaking underground storage tank releasing oil into a river
27. Which of the following is not a potential challenge in implementing integrated watershed management?
- a. Limited funding and resources
 - b. Conflicting interests among stakeholders
 - c. Lack of data and information about the watershed
 - d. **Existing** water quality standards
28. What is the purpose of a watershed assessment in integrated watershed management?
- a. To identify and prioritize watershed issues and needs
 - b. To develop specific management actions to address those needs
 - c. To evaluate the effectiveness of management actions
 - d. All **of the** above

29. What is the main purpose of public participation in integrated watershed management?
- To ensure **that** decisions reflect the values and needs of the community
 - To hold stakeholders accountable for their actions in the watershed
 - To limit the involvement of the public in decision-making
 - To increase conflict among stakeholders
30. These are made from locally available materials and have a life up to 8 years
- Temporary gully control structure
 - Permanent **gully** control structure
 - Drop spillway
 - Chute spillway
31. The following are the basic types of permanent gully control structure Except
- Drop inlet spillway
 - Drop spillway
 - Woven **wire** check dam
 - Chute spill way
32. Gullies are formed by an increase in
- Precipitation
 - Water** erosion
 - Wind erosion
 - Surface runoff
33. Most of the time the major problems occurred in the highland parts of Ethiopia that affects soil productivity is;
- Sodicity
 - Salinity
 - Alkalinity
 - d. Acidity**
34. Essential plant nutrients having less attention in soil fertility management are _____
- C, H, O
 - N, P, K
 - Ca, Mg, S
 - d. Fe**, Cu, Zn
35. The amount of potentially available nutrient that can be changed to available form through time indicates;

- a. Intensity
 - b. Quantity
 - c. Buffer power
 - d. Buffer capacity
36. A term that indicates a situation in which a crop needs of a given element, yet has shown no deficiency symptom is;
- a. Luxury consumption
 - b. Toxicity
 - c. Hidden hunger
 - d. sufficiency stage
37. Which one of the following forms of nitrogen is directly available for plant uptake?
- a. Ammonium
 - b. Nitrite
 - c. Nitrogen gas
 - d. All
38. Which one of the following is the function of phosphorus?
- a. Energy transfer reactions
 - b. Crop maturity
 - c. Development of reproductive structures
 - d. All
 - e.
39. Which one of the following method/technique could be important to determine periodic change of nutrients with in the soil?
- a. Nutrient-deficiency symptoms of plant
 - b. Plant analysis
 - c. Microbiological method
 - d. Field test
40. Which one of the following essential element increases with organic matter addition?
- a. Boron
 - b. Iron
 - c. Manganese
 - d. All
41. Which one of the following process reduces the availability of nutrients?
- a. Fertilization

- b. Dissolution
- c. Mineralization
- d. Leaching

42. From the following which one is not considered as the key features of ISFM principles?

- a. Replenishing soil nutrient pools
- b. Minimizing on-farm recycling of nutrients
- c. Reducing nutrient losses to the environment
- d. Improving the efficiency of external inputs

43. Impact analysis stage that has the objective to estimate the magnitude, extent and duration of impact in comparison with the situation and without that project.

- a. Impact evaluation
- b. Impact identification
- c. Impact prediction
- d. Impact significance

44. Which methods of evaluating impact significance has the basic aim to assist the decision maker in electing alternative from a number of feasible choice alternatives?

- a. Matrices technique
- b. Multiple criteria decision making
- c. Cost benefit analysis
- d. Using environmental standards

45. The monitoring process that involves the measurement of performance indicators during project construction and implementation to detect changes occurred as a result of project?

- a. Impact/effect monitoring
- b. Base line monitoring
- c. Compliance monitoring
- d. Mitigation monitoring

46. At which stages of the EIA process that the public as a stakeholder participated when they involved in the time of identification of issues and alternatives to be considered?

- a. During screening
- b. During scoping
- c. During impact mitigation
- d. During impact identification

47. If you have the position to review the ESIA document, which one of the following is less relevant during your evaluation time of the document?
- The terms of reference
 - The **owner** of the project
 - The severity of the impacts and the possible mitigation methods,
 - The potential benefits of the projects
48. Which one of the following is not a component of a good EIA report and review?
- Impact assessment, mitigation measures, and follow-up plans
 - A terms of reference
 - A **comprehensive** set of descriptions about the project benefits described using highly technical terms.
 - A proper and adequate prediction of the adverse impacts of proposed projects and their mitigation using conventional and customized techniques.
49. Which proclamation number in Ethiopia has made environmental impact assessment to be a mandatory legal requirement for major development projects?
- Proc. No 295/2002
 - Pro. No 300/2002
 - proc.** No 299/2002
 - Article number 43 and 44 of the FDRE constitution
50. Which one of the following is not best describes the executive summary of the EIA report?
- Gives a **detailed** description of the main findings and recommendations.
 - Should be kept short and concise
 - Should be clearly written and free from jargon and technical words
 - It should target to the audience
51. A core part of impact management is developing an environmental management plan. Please select the one that is not the content of the environmental management plan?
- Mitigation
 - Implementation Schedule and Cost Estimates
 - Capacity Development
 - Terms** of references
52. Select the correct relation between types of uncertainty and the ways in which the impact can be minimized
- uncertainty of prediction minimized by coordination

- b. decision related uncertainty minimized by research
 - c. Uncertainty of values minimized by research
 - d. **Uncertainty** of values minimized by improved communication
53. Which strategy is commonly used to manage the environmental impact of energy consumption?
- a. Increasing fossil fuel production
 - b. **Implementing** carbon pricing
 - c. Reducing public transportation options
 - d. Encouraging deforestation
54. What is the primary goal of sustainable energy production and per capita consumption?
- a. **Minimize** energy use, waste production, and emissions throughout the entire lifecycle
 - b. Maximize energy use, waste production, and emissions throughout the entire lifecycle
 - c. Decrease energy use while increasing emissions throughout the entire lifecycle
 - d. Increase energy use and waste production throughout the entire lifecycle
55. Which one is the main advantage of using biofuels over fossil fuels?
- a. Biofuels are cheaper to produce
 - b. Biofuels are **renewable** and sustainable
 - c. Biofuels are more emissions
 - d. Biofuels are stringent environmental pollution
56. Which one is the ultimate source of energy?
- a. Uranium
 - b. Hydropower energy
 - c. **Electromagnetic** radiation
 - d. Fossil fuels
57. Which one of the following forms of energy leads to stringent environmental pollution in the process of its harnessing and utilization?
- a. **Fossil** fuel energy
 - b. Wind energy
 - c. Solar energy
 - d. Bio energy

58. Why is there a demand for renewable energy sources than none renewable energy sources?
- Because they are less energy efficient
 - Because they emit more greenhouse gases
 - Because of the increasing global temperatures
 - Because** of low or zero carbon footprints
59. Which of the following technologies converts sunlight directly into electrical energy?
- Photovoltaic** (PV) panels
 - Solar thermal power plants
 - Wind turbines
 - Hydroelectric power plants
60. Which of the following technologies converts the chemical energy stored in organic materials into electrical or thermal energy?
- Biomass** power plants
 - Fossil fuel power plants
 - Solar thermal power plants
 - Hydropower plants
61. Which process is utilized to generate biogas from organic waste?
- Fermentation
 - Gasification
 - Anaerobic** digestion
 - Trans-esterification
 -
62. What is the primary goal of energy efficiency programs?
- To increase energy consumption
 - To reduce the cost of energy production
 - To minimize the amount of **energy used to provide** the same service
 - To eliminate the use of fossil fuels
63. Which one of the following statements is correct about the importance of genetic diversity?
- Genetic diversity plays an important role in the survival and adaptability of a species
 - The resistance of the individuals of a population to certain types of diseases can increase with the increase in genetic diversity.

- c. Genetic diversity is essential for a species to evolve.
- d. All of the above.

64. Which one of the following statements is incorrect about a flagship species?

- a. A flagship species is a species from only the animal kingdom that is used to represent a certain environmental issue or cause.
- b. The use of flagship species can skew the management and conservation priorities in that they favor and in-detriment of more threatened species.
- c. The managements of different flagships can conflict.
- d. The disappearance of the flagship can have negative impacts on the attitudes of the conservation stakeholders.

65. Which one of the followings is false about keystone species?

- a. It is an animal or plant species that has an important ecological impact much greater than its abundance would otherwise suggest.
- b. It is a species that plays a critical role in maintaining the structure and functions of an ecological community.
- c. It affects many other organisms in an ecosystem and helps to determine the types and numbers of various other species in a biotic community.
- d. None of the above.

66. Which one of the following statements is correct about umbrella species?

- a. Umbrella species refers to only large mammals, since they tend to have the greatest range of environments and often have a large impact on their ecosystem.
- b. Umbrella species has a wide range that has requirements for living that are commonly higher than other associated species in its habitat.
- c. The use of umbrella species is designed to make the conservation and environmental decision-making process more difficult.
- d. The size of the reserve to protect and conserve the biodiversity is always less than the size of the ranges of the umbrella species.

67. Which one of the following indicator species whose presence indicates low air pollution in the terrestrial ecosystem?

- a. Stoneflies
- b. Lichens
- c. Greasewood
- d. Mosses

68. Which one of the following processes regulates the patterns of biodiversity?
- a. History: Correlation through time as a function of reproduction.
 - b. Chance: Random processes of birth, death, and movement (i.e., migration, emigration, and /or immigration).
 - c. Necessity: The laws of growth, competition, predation, and interaction.
 - d. All of the **above**.
69. One of the following statements is not correct about stratified sampling?
- a. Stratified sampling is applicable when the **population** cannot be exhaustively partitioned into disjoint subgroups.
 - b. The strata should be collectively exhaustive.
 - c. The strata should be mutually exclusive.
 - d. Proportionate allocation uses a sampling fraction in each of the strata that is proportional to that of the total population.
70. Which one of the followings is incorrect about a foundation species?
- a. It has been applied to range of organisms in ecosystems around the world, in both aquatic and terrestrial environments.
 - b. It can only occupy a **producer** any trophic level in a food web.
 - c. It refers to a species that has a strong role and influence in structuring a biotic community.
 - d. The activities of foundation species physically modify the environment, produce and maintain habitats that benefit other organisms that use those habitats.
71. Which one of the following is true about biodiversity indicators?
- a. They help understand the consequences of biodiversity changes on ecosystems, their services, and human well-being.
 - b. They use quantitative data to measure aspects of biodiversity, ecosystem condition, services, and drivers of change.
 - c. They are statistical measures of biodiversity which help scientists, managers, and politicians to understand the condition of biodiversity and the factors that affect it.
 - d. All of **the** above.
72. Which one of the following statements is false?

- a. In floristic data collection, it is important that the same person should do the estimations of the plant cover in each quadrat, it is because the estimation is likely to vary from person to person.
- b. In locating the sample plots along the transect, the length of the sub-divisions or the distance between sample plots depends on the pattern of environmental change.
- c. In locating sample plots along the transect, when there is a rapid environmental change, fewer sub-divisions are required, but when there is slow environmental change, more sub-divisions are required.
- d. In floristic vegetation description method, quadrat size is affected by the vegetation type.

73. Which one of the following statements is incorrect?

- a. A limiting factor is any factor that places an upper limit on the size of a wildlife population.
- b. Limiting factors may be biotic, such as the availability of food, or abiotic, such as access to water.
- c. Human influences often also act as limiting factors.
- d. None of the above.

74. A place of refuge where abused, injured and abandoned captive wildlife may live in peace and dignity for the remainder of its life is called _____.

- a. Wildlife Sanctuary
- b. Wildlife Reserve
- c. Wildlife National Park
- d. Wildlife-Controlled Hunting Area

75. Which one of the followings is a common idea that should be present in every definition of a wildlife management?

- a. Efforts directed towards wild animal populations.
- b. Relationship of habitat to those wild animal populations.
- c. Manipulations of habitats or populations that are done to meet some specified human goal.
- d. All of the above.

76. Which one of the followings is incorrect about capture-recapture method?

- a. Mortality is the same for marked and unmarked animals.
- b. Marked animals are mixed randomly with unmarked animals.
- c. The population has **significant** ingress (births or immigration) or egress (deaths or emigration).
- d. Marked individuals are caught at the same rate as unmarked individuals (no trap-happy or trap-shy animals).

77. Which one of the followings is not included in the goals of active wildlifemanagement?

- a. Make wildlife population increase through ensuring wildlife survival.
- b. Leave wildlife **population** alone, but keep an eye on it.
- c. Make wildlife population decrease to control over abundance.
- d. Harvest wildlife population for a sustained yield.

78. During the process of the development of commercial breeding of wildlife species (e.g., through rearing and farming methods), which one of the following points should be considered for controlled trade?

- a. Develop a system whereby there is a net gain to wildlife population.
- b. Link farming and utilization closely to adequate protection of wild population.
- c. Development a control system that makes it difficult for wild stock to be poached and traded as farmed or domestic animals.
- d. All of the **above**.

79. Wildlife foods that are eaten on a regular basis to meet the nutritional needs of the wild animals are called _____.

- a. Preferred foods
- b. Staple **foods**
- c. Emergency foods
- d. Starvation foods

80. Which one of the following statements is correct about wildlife dispersal barriers?

- a. A dispersal barrier may mean that the dispersal range of a species is much smaller than the species natural distribution.

- b. An artificial example of wildlife barrier is habitat fragmentation created due to human land use, such as deforestation, road construction, and human settlements.
- c. Natural barriers to dispersal that limit species distribution include mountain ranges and rivers.
- d. All of the above.

81. Which one of the followings is not an example of non-consumptive wildlife utilization in Ethiopia?

- a. Wildlife-based tourism in national parks.
- b. Civet musk collection from individually reared animals.
- c. Sport-hunting at different Wildlife-Controlled Hunting Areas.
- d. Incense and gums collection from arid and semi-arid parts of Ethiopia.

82. Which one of the followings should be considered to meet the assumptions related to the use of point transects / counts?

- a. It requires the observer to remain at one point for a fixed time and recording all the close animals detected.
- b. Individuals being sampled are detected at their initial location prior to any movement in response to the observer.
- c. The recording period must be long enough for all close animals to the observer to be recorded.

d. All of the above.

83. One of the following is concerned with putting environmental resources to new kinds of productive use

- a. Land evaluation
- b. Land use cover
- c. Land use planning
- d. Land use type

84. Which one of the following reflects degree of suitability of the land?

- a. Order
- b. Classes
- c. Unit
- d. Subclasses

85. Which one indicates comprehensive evaluation involving in-depth analysis of land characteristics?
- a. Detailed **assessment**
 - b. Semi-Detailed assessment
 - c. Rapid assessment
 - d. Qualitative approach
86. _____ is a powerful tool for land evaluation and land cover classification.
- a. Agricultural data
 - b. Climate data
 - c. Soil survey
 - d. Remote **sensing**
87. Accepting the diverse functions of land ownership helps to address issues of
- a. Equity
 - b. Sustainability
 - c. Economic Development
 - d. All of the **above**
88. One of the following is not the use of land use planning?
- a. For urban and regional development
 - b. For short-term **sustainability**
 - c. For sustainable agriculture and food security
 - d. For environmental protection
89. A common individual objectives in land use planning that concerned on economic objectives only
- a. Public engagement
 - b. Conservation and sustainability of land
 - c. Encouraging business **development**
 - d. Resource management
90. One of the following is level of land use planning that influence national and regional planning frameworks by setting global objectives on issues like climate change, biodiversity, and sustainable land use
- a. Local level
 - b. Site specific level

- c. Regional level
- d. National level

91. Why do you think that land use planning is flexible?

- a. For dis-couraging innovation
- b. For non-adaptability to change
- c. For improved risk management
- d. For focus on dis-continuous improvement

92. Which one of the following is refer to the continuous observation and recording of land-use activities to track their progress, effectiveness, and compliance with the land-use plan?

- a. Data collection
- b. Monitoring
- c. Evaluation
- d. Upgrading the land-use plan